Investigating Efficient Learning Algorithms for Cooperative Task

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Abstract

This thesis investigates efficiency in cooperative Multi-Agent Reinforcement Learning (MARL), focusing on the design of agent models, reward structures, and sequential decision processes. Cooperative MARL enables multiple agents to coordinate actions toward shared objectives, often outperforming single-agent approaches. However, extending reinforcement learning to multi-agent settings introduces challenges, including non-stationarity, partial observability, credit assignment, and the combinatorial growth of state-action spaces. Addressing these challenges requires careful modelling of agent observations, interactions, and task-specific rewards.

The theoretical component of this work reviews state-of-the-art MARL methods, highlighting strategies that reduce complexity and improve scalability. Two main paradigms are identified: centralised approaches, which model agents as a single joint entity, and decentralised approaches, in which agents act independently based on local observations. Across both paradigms, effective methods exploit sparsity to remove redundancy and leverage composition or invariants to simplify representations. Based on these insights, hypotheses are formulated along three key dimensions: (i) decision processes (sequential versus parallel coordination), (ii) state-space modelling (centralised versus decentralised observations), and (iii) reward design (shaped versus sparse rewards, e.g., via potential fields).

The experimental component tests these hypotheses using the Multi-Agent Path Finding (MAPF) task. A composite method is proposed that integrates potential-based reward shaping, shared episodic memory, and information-theoretic exploration under a Centralised Training, Decentralised Execution (CTDE) paradigm. Comparative experiments against a baseline Independent Q-Learning (IQL) framework demonstrate that the composite method improves convergence speed, sample efficiency, and coordination among agents. Visual analyses of potential fields, agent trajectories, and joint value functions provide interpretability and insights into emergent cooperation.

Overall, this thesis demonstrates that combining theoretically informed modelling choices—reward shaping, memory mechanisms, and structured exploration—can significantly enhance training efficiency in cooperative MARL. The findings provide a systematic framework for designing efficient multi-agent learning algorithms and highlight practical strategies for scalable, robust coordination in complex environments.

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I certify that this project is my own work. Code development and debugging were assisted by ChatGPT 4.0. The report structure and initial drafts are my own; final editing was assisted by ChatGPT for clarity and consistency. All outputs have been reviewed and edited to accurately reflect my work.

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1. Introduction

The efficiency of training in cooperative multi-agent systems remains a central challenge in reinforcement learning research. While reinforcement learning (RL) has demonstrated remarkable potential, it is often criticised for its high training time, data inefficiency, and substantial computational demands [Zhang et al. 2024]. These limitations become even more pronounced in **Multi-Agent Reinforcement Learning (MARL)**, where the interactions between multiple agents introduce additional complexity. This thesis explores strategies to improve efficiency in cooperative MARL by examining modelling choices, reward structures, and decision-making frameworks, with the aim of advancing both theoretical understanding and practical applicability.

1.1 Motivation

The study of **Cooperative Multi-Agent Reinforcement Learning (MARL)** is motivated by its potential to model and analyse complex systems involving multiple decision-makers. From a computational perspective, cooperative MARL offers scalable approaches to problems that are otherwise intractable due to high-dimensional state-action spaces. From a prescriptive perspective, it provides a framework for designing policies that facilitate coordination and maximise collective performance. From a descriptive perspective, it serves as a powerful tool for understanding and replicating cooperative behaviours observed in both natural and artificial systems [Tan, 1993].

Despite this potential, a major limitation of reinforcement learning, and by extension MARL, is its inefficiency in terms of sample requirements, training time, and computational cost [reference]. These inefficiencies are further exacerbated in cooperative settings, where the interaction of multiple agents introduces additional complexity, including non-stationarity, credit assignment, and scalability challenges. Addressing these issues is crucial for developing MARL algorithms that are not only theoretically robust but also practically deployable.

The importance of efficiency is underscored by the demands of real-world applications. Domains such as **autonomous driving**, **robotic swarms**, **telecommunication networks**, and **distributed energy systems** all involve inherently cooperative tasks in which agents must coordinate their actions to achieve shared objectives. In such environments, effective cooperation is essential for ensuring safety, reliability, and performance. However, the practical deployment of MARL in these settings is constrained by the inefficiency of current algorithms. Excessive data requirements, slow convergence, and high computational costs impose barriers to scalability and real-time applicability.

Thus, while cooperation is the defining feature of MARL, **efficiency constitutes a critical enabling factor** that determines its relevance beyond theoretical research. Enhancing efficiency not only expands the applicability of cooperative MARL to increasingly complex domains but also ensures that its benefits can be realised in practice. This thesis addresses this challenge by investigating the causes of inefficiency in cooperative MARL and exploring strategies to improve learning performance, both theoretically and experimentally.

1.2 Objectives

The objective of this thesis is to investigate how efficiency in **Cooperative Multi-Agent Reinforcement Learning (MARL)** can be improved through both theoretical and empirical inquiry. On the theoretical side, the work seeks to develop general insights into the factors that influence efficiency, with particular attention to modelling strategies, reward structures, and decision-making paradigms. This theoretical analysis serves as a foundation for generating hypotheses regarding the mechanisms that underpin efficient learning in cooperative settings. Complementing this, the thesis undertakes an experimental investigation designed to test these hypotheses in practice. By evaluating efficiency-enhancing methods within cooperative MARL environments, both individually and in combination as a composite approach, the research aims to bridge conceptual understanding with empirical evidence, thereby advancing both theoretical and practical knowledge of efficiency in cooperative MARL.

1.3 Contributions

This research makes several contributions to the study of efficiency in Cooperative Multi-Agent Reinforcement Learning (MARL). First, it provides a synthesis of existing successes in improving efficiency within cooperative tasks, offering a structured perspective on the state of the art. Building on this analysis, the work generates and evaluates hypotheses concerning the factors that enable more efficient learning in cooperative MARL.

To the best of my knowledge, the combination of methods explored in this thesis, specifically **reward shaping via artificial potential fields**, **shared state-space history**, and **curiosity-driven or intrinsic motivation mechanisms**, has not previously been studied as a unified approach. This integrated strategy, referred to in this thesis as a **composite method**, represents a novel contribution to the field. By experimentally testing this composite method, the thesis advances understanding of how diverse efficiency-enhancing techniques can be jointly applied to cooperative MARL.

2. Theory

2.1 Reinforcement Learning

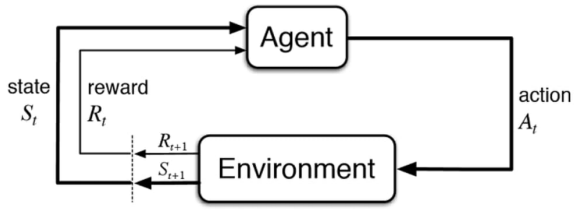

Figure 1: Standard RL Model

2.1.1 Formal Model

Reinforcement Learning (RL) provides a computational framework for sequential decision-making under uncertainty, in which an agent interacts with an environment by selecting actions, receives feedback in the form of rewards, and iteratively learns to maximise cumulative reward [Sutton & Barto, 2018]. The standard formalism for RL is the **Markov Decision Process (MDP)** [Puterman, 1994], which is typically defined by a tuple (S, A, P, R, γ), where:

- S is a finite set of states.
- A is a finite set of actions.
- $P(s'|s,a)$ denotes the transition probability function
- $R(s,a)$ denotes the reward function
- $\gamma \in [0,1]$ is the discount factor, determining the relative importance of future versus immediate rewards.

In many practical scenarios, the agent does not have full access to the true state of the environment, giving rise to a **Partially Observable Markov Decision Process (POMDP)**, where the agent instead receives observations that provide incomplete information about the underlying state. The objective of RL in this context is to learn an optimal policy π(a | s) that maximises the expected discounted cumulative reward over time. Solving the POMDP in this sense is considered the fundamental goal of reinforcement learning.

2.1.2 Components of Model

The development of a cooperative multi-agent reinforcement learning model can be conceptualized through three interconnected components: the **agent model**, the **target model (environment)**, and the **sequential decision process model**.

The **agent model** formalizes the internal representation of each learning agent, including its states and the dynamics governing state transitions. By capturing the agent's perception of the environment and its internal decision-making mechanisms, this model enables the agent to select actions that influence both its own trajectory and the outcomes of other agents in the system.

The **target model**, representing the environment, characterizes the mapping from agent actions to outcomes. This includes the value functions and reward structures that define how

agent behaviours are evaluated, as well as the value transition dynamics that determine how the environment evolves in response to agent interactions. Together, these components define the feedback loop through which agents learn and adapt.

Finally, the **sequence and decision process model** formalizes the temporal and probabilistic structure of interactions. By assuming the Markov property, where the future state depends only on the current state and chosen action—this component provides the theoretical foundation for modelling sequential decision-making in both fully and partially observable settings. It underpins the formulation of policies and the computation of expected cumulative rewards, serving as the bridge between agent behaviours and environmental outcomes.

Collectively, these three components form a structured framework for constructing cooperative MARL systems, enabling systematic analysis, simulation, and evaluation of learning efficiency.

2.1.3 Algorithmic Paradigms

Reinforcement learning (RL) algorithms can be conceptualized as the combination of a **sequential decision model**, which captures the temporal evolution of states and actions, and a **policy update mechanism**, typically expressed through variants of the Bellman equation. These algorithms iteratively refine an agent's policy to maximise the expected cumulative reward over time.

2.1.3.1 Assumptions Underlying Learning Algorithms

RL algorithms rely on explicit assumptions regarding the knowledge available to the agent:

- **Known components**: These may include the set of possible states, actions, and the structure of the reward function.
- **Unknown components**: The agent typically has incomplete information about transition dynamics or reward outcomes, which must be learned through interaction with the environment.

The agent's learning process is structured as sequences of **episodes** and **time steps**, providing a framework for modelling sequential decision-making in both fully and partially observable environments.

2.1.3.2 Algorithmic Families

Reinforcement learning methods are generally categorized into two primary families: **Dynamic Programming (DP)** and **Temporal Difference (TD) Learning**. Both families utilize the Bellman equation to iteratively update state or action values, but they differ in their assumptions and operational requirements.

Dynamic Programming methods assume complete knowledge of the environment's transition probabilities and reward function. The Bellman equations provide a recursive definition of the value of a state in terms of the expected value of its successor states. Classical DP approaches, such as **value iteration** and **policy iteration**, exploit this structure to converge to an optimal policy under the assumption of a fully specified model.

Temporal Difference Learning relaxes the requirement of a complete model by allowing agents to learn directly from sampled experiences. TD methods, including **SARSA** and **Q-learning**, update value estimates using observed transitions and received rewards. By combining aspects of Monte Carlo sampling and DP, TD learning provides a practical framework for learning in environments where transition probabilities are unknown or costly to compute.

2.1.3.3 Fundamental Components of Algorithms

Both DP and TD paradigms share several key components that define the learning process:

- **Bellman Equation**: Expresses the relationship between immediate reward, discounted future rewards, and state transitions.
- **Explore-Exploit Trade-off**: Determines the balance between exploring unknown actions and exploiting known high-value actions.
- **Discount Factor (γ)**: Governs the relative weighting of future versus immediate rewards.
- **Learning Objective**: Formulated as the maximization of the expected cumulative discounted reward.
- **Update Mechanism**: Specifies how value functions or policies are adjusted based on observed experience or model knowledge.

Understanding these paradigms is essential for analysing the efficiency, convergence properties, and scalability of reinforcement learning algorithms, particularly in the context of cooperative multi-agent systems.

2.1.4 Solution Concepts

In reinforcement learning, the **solution concept** formalizes the objective of the learning process by defining what constitutes an optimal policy. The learning objective is typically to identify a policy π^* that maximizes the expected cumulative reward over time, accounting for both immediate and future returns. Formally, for a (PO)MDP with state space S, action space A, reward function R, transition dynamics P, and discount factor $\gamma \in [0,1]$, the optimal policy is defined as:

$$\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \,\middle|\, \pi \right]$$

where the expectation is taken over the stochastic transitions induced by the environment and the agent's policy.

In multi-agent settings, particularly cooperative MARL, the solution concept extends to joint policies $\pi* = (\pi 1*, \ldots, \pi N^*)$ that maximize a **collective objective**, such as the total expected cumulative reward across all agents. This requires accounting for interactions among agents, partial observability, and potential non-stationarity of the environment due to simultaneous learning.

The solution concept thus provides a formal target for learning algorithms, guiding the design of policy update mechanisms, value function approximations, and exploration strategies, and

serving as the foundation for evaluating the efficiency and effectiveness of cooperative MARL methods.

2.2 Game Theory

2.2.1 Modelling Components

A game-theoretic model provides a formal representation of strategic interactions among multiple decision-making agents. At its core, a **game** is defined by a set of agents, the possible actions or strategies available to each agent, and the payoff structure that determines the outcomes of these strategies [von Neumann, 1944]. Establishing consistent terminology is essential for linking classical game theory concepts to multi-agent reinforcement learning frameworks.

The primary components of a game model include:

1. **Set of Agents**: The individual decision-makers whose actions collectively determine the outcome of the game.
2. **Target (Payoff)**: The objective or reward structure for each agent. Depending on the interaction type, games can be classified as **competitive**, **cooperative**, or **mixed**. In cooperative settings, the payoff often reflects collective performance, aligning the agents' objectives toward shared goals.
3. **Sequential and Decision Process**: The temporal structure of the game, which captures the order of decisions, the evolution of states, and how agent actions influence both immediate and future payoffs. This is analogous to the sequential decision processes formalized in (PO)MDPs and MARL.

2.2.2 Solution Concepts

Solution concepts formalize what it means for agents to act optimally in a game-theoretic context. They provide a means to study **optimal strategies** for given game models, accounting for the strategic interdependence of agents. Classical solution concepts, such as **Nash equilibrium**, define strategy profiles in which no agent can unilaterally improve its payoff by deviating from its chosen strategy. In cooperative MARL, these concepts are extended to joint policies that maximize collective rewards, allowing for a principled framework to evaluate the effectiveness of learning algorithms and the efficiency of coordinated behaviours.

By connecting the formal components of game models with MARL frameworks, this approach enables rigorous analysis of multi-agent interactions, the design of efficient learning algorithms, and the interpretation of agent behaviours through equilibria in both competitive and cooperative domains.

2.3 Multi-Agent Reinforcement Learning

Multi-Agent Reinforcement Learning combines **reinforcement learning**, which enables agents to learn optimal policies through interaction, with **game theory**, which models

strategic interactions among multiple agents. This synthesis allows agents to learn policies that account for others' actions, generalizing single-agent RL to multi-agent environments.

2.3.1 Formal Multi-Agent Models

In multi-agent systems, interactions among N agents are commonly modelled as **stochastic games**, generalizing the single-agent Markov Decision Process (MDP). Formally, a stochastic game is defined as a tuple:

$$\mathcal{G} = \langle \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, P, \{R_i\}_{i=1}^N, \gamma \rangle,$$

where:

- $\mathcal{S}$ is the set of environment states,
- $\mathcal{A}_i$ is the set of actions available to agent i,
- $P : \mathcal{S} \times \mathcal{A}_1 \times \cdots \times \mathcal{A}_N \times \mathcal{S} \to [0, 1]$ is the state transition probability,
- $R_i : \mathcal{S} \times \mathcal{A}_1 \times \cdots \times \mathcal{A}_N \to \mathbb{R}$ is the reward function for agent i,
- $\gamma \in [0, 1]$ is the discount factor.

When agents have **partial observability**, each agent iii receives observations oi ∈ Oi according to an observation function:

$$O_i : \mathcal{S} \times \mathcal{A}_1 \times \cdots \times \mathcal{A}_N \to \mathcal{O}_i.$$

Such systems are referred to as **Partially Observable Stochastic Games (POSGs)**, or **Dec-POMDPs** in decentralized settings.

2.3.1.1 Centralized Multi-Agent Models

In the **centralized paradigm**, all agents are treated jointly with access to the global state, s ∈ S. The joint action is defined as:

$$\mathbf{a} = (a_1, a_2, \ldots, a_N) \in \mathcal{A}_1 \times \cdots \times \mathcal{A}_N.$$

The centralized stochastic game then reduces to a **Multi-Agent MDP (fully observable)** or a **Multi-Agent POMDP (partially observable)** with a joint reward function R: S × A → R. Optimal joint policies π∗ = (π1∗, …, πN*) maximize the expected cumulative return:

$$\pi^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \mathbf{a}_t)\right].$$

While theoretically optimal, centralized formulations suffer from **combinatorial explosion** in |A1×⋯×AN|.

2.3.1.2 Decentralised Multi-Agent Models

In the **decentralized paradigm**, each agent iii observes only its local observation oi and chooses actions according to a local policy πi: Oi →Ai . The joint policy is π = (π1, …, π*), and each agent aims to maximize its expected return:

$$J_i(\pi) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_i(s_t, \mathbf{a}_t) \mid \pi\right].$$

This formalism gives rise to **Decentralized MDPs (Dec-MDPs)** or **Dec-POMDPs**, which are computationally tractable for large-scale systems and reflect practical constraints, such as limited communication and partial state observability [Oliehoek & Amato, 2016; Bernstein et al., 2002].

3. Literature Review

This literature review analyses the formal models of Multi-Agent Reinforcement Learning (MARL) introduced in the preceding theoretical chapter, breaking them down into their constituent components. This component-based interpretation provides a structured framework for evaluating existing research, highlighting the modelling approaches, the principal challenges associated with these approaches, and contemporary solutions proposed to address them.

To guide the review, three central questions are posed: (1) How can agents be modelled most effectively? (2) How can targets and objectives be optimally represented? (3) How can the sequential decision process be formulated to maximise learning efficiency? While the review primarily focuses on value-based methods, policy-gradient approaches are acknowledged as relevant but are not the focus of this analysis.

By framing the literature around these questions and component-based challenges, the review establishes a coherent foundation for identifying research gaps and motivates the subsequent experimental investigations into improving efficiency in cooperative MARL tasks.

3.1 How to Model Agents?

3.1.1 Hierarchical Interpretation of Modelling Multi-Agent Systems

It is instructive to conceptualize multi-agent systems within a **two-level hierarchical framework** [Xu et al., 2022], as this perspective sharpens the distinction between centralized and decentralized modelling paradigms. At the highest level, the centralized formulation, commonly formalised through stochastic games, assumes a single controller with access to the global system state, making joint decisions on behalf of all agents. In contrast, the decentralized formulation, formalised in decentralized partially observable Markov decision processes (Dec-POMDPs) or implemented through independent learners, restricts each agent to acting based on its own local observations.

Crucially, the hierarchical view highlights the interdependence of these paradigms. A centralized agent can be regarded as the composition of all decentralized agents into a single global decision-maker, while decentralized agents can be understood as the decomposition of a centralized controller into distributed decision-makers with partial autonomy and limited observability. This composition–decomposition relationship is not only conceptual but also methodological, since many MARL algorithms occupy positions along a spectrum between fully centralized and fully decentralized approaches.

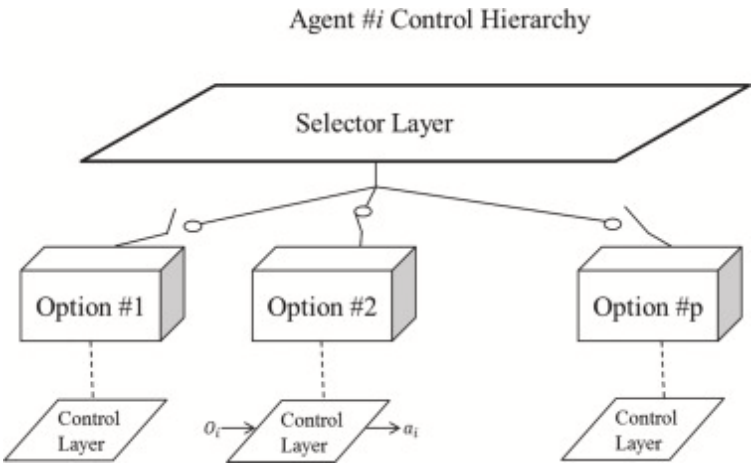

Figure 2: Hierarchical Interpretation

3.1.2 Observability in Agent Models

A fundamental dimension in the hierarchical modelling of multi-agent systems is the concept of **observability**, which defines the information available to an agent when making decisions. In the case of full observability, as assumed in centralised models, the agent, or a central controller, has complete access to the global state of the environment. This implies that the transition dynamics and reward functions are entirely known from the agent's perspective, allowing standard reinforcement learning methods, such as Markov Decision Processes (MDPs), to be applied directly. Full observability simplifies policy learning by removing uncertainty about the system's current state, ensuring that optimal actions can, in principle, be determined with respect to the true environment dynamics.

In contrast, partial observability is characteristic of decentralised models, where each agent perceives only a subset of the full state. The observation function may vary between agents and across time, depending on local sensors, environmental occlusions, or limited communication. From a reinforcement learning perspective, partial observability transforms the problem into a **Partially Observable Markov Decision Process (POMDP)**, or in multi-agent settings, a **Decentralised POMDP (Dec-POMDP)**. Under partial observability, an agent cannot directly infer the full state and must rely on histories of observations, inferred hidden states, or communication with other agents to approximate the true environment state. This introduces significant challenges in learning stable and coordinated policies, as the agent's perceived environment is inherently non-stationary and stochastic, particularly in the presence of other adaptive agents.

Partial observability thus provides a unifying perspective on centralised and decentralised control. Centralised models can be seen as the limiting case of complete information, whereas decentralised models represent the generalised scenario of distributed, incomplete information. Recognising this continuum is essential for understanding the trade-offs between computational tractability, coordination efficiency, and robustness in multi-agent reinforcement learning. Moreover, it motivates the design of algorithms that exploit available information optimally, whether through memory mechanisms, communication protocols, or structured inference over hidden states.

3.1.3 Review on Centralised Interpretations

3.1.3.1 Centralised Agent Models

The centralised model of agents conceptualises a multi-agent system as if it were governed by a single global decision-maker, or *central agent*, endowed with access to the complete state of the environment. In this formulation, the collection of n agents is represented as a composite entity, thereby reducing the problem to a unified reinforcement learning setting. The single-agent case arises naturally when n=1, in which the framework collapses to the standard Markov Decision Process (MDP). For n>1, the central agent is assumed to control a joint policy across all constituent agents, operating over the joint state space and joint action space induced by their individual contributions.

A defining feature of this paradigm is the distinction between homogeneous and heterogeneous agents. Homogeneous agents share identical observation and action spaces, rendering their interactions structurally equivalent. For example, in spatial–temporal environments, all agents may perceive the same grid structure, albeit from different local perspectives. By contrast, heterogeneous agents differ in their observation or action spaces, reflecting asymmetries in roles, capabilities, or access to information. This heterogeneity introduces additional complexity, as the joint policy must coordinate diverse behaviours while accommodating fundamentally different modes of interaction with the environment.

The centralised perspective offers conceptual clarity by unifying single-agent and multi-agent settings within a common framework. It provides a structured lens for analysing cooperative and competitive interactions, particularly under the idealised assumptions of perfect communication and seamless integration of local observations into a global state. However, these idealisations expose methodological challenges. The joint state–action space expands exponentially with the number of agents, rendering exact learning intractable for all but the smallest systems. Moreover, the assumptions of full observability and unconstrained communication are rarely met in practice, where agents often face information bottlenecks or incomplete state access.

As such, centralised models are most valuable as theoretical baselines or conceptual reference points. They combine simplicity and interpretability but are typically impractical in large-scale applications, motivating the search for hybrid approaches that balance the informational advantages of centralisation with the scalability of decentralised decision-making.

3.1.3.2 Baseline Model for the Centralised Case

A widely adopted baseline in this setting is **Centralised (or Joint) Q-Learning** [reference], formulated within the framework of stochastic games. Here, the joint action space of all agents is treated as a single unified decision space, and the learning objective is to estimate a joint action–value function that guides the selection of globally optimal joint policies. This baseline has appeal due to its theoretical clarity: it provides a straightforward extension of single-agent Q-Learning and serves as a natural benchmark against which more sophisticated methods can be evaluated.

1: Initialize: $Q(s,a)=0$ for all $s \in S$ and $a \in A = A_1 \times \ldots \times A_n$
2: Repeat for every episode:
3: **for** $t = 0, 1, 2, \ldots$ **do**
4: Observe current state s^t
5: With probability ϵ: choose random joint action $a^t \in A$
6: Otherwise: choose joint action $a^t \in \arg\max_a Q(s^t, a)$
7: Apply joint action a^t, observe rewards $r_1^t, \ldots, r_n^t$ and next state s^{t+1}
8: Transform $r_1^t, \ldots, r_n^t$ into scalar reward r^t
9: $Q(s^t, a^t) \leftarrow Q(s^t, a^t) + \alpha \left[r^t + \gamma \max_{a'} Q(s^{t+1}, a') - Q(s^t, a^t) \right]$

Figure 3: Algorithm 1 Central Q-Learning (CQL) for stochastic games

3.1.4 Limitations of the Centralised Baseline

Despite its role as a reference point, Centralised Q-Learning faces well-documented limitations. The joint state–action space grows exponentially with the number of agents, creating severe scalability challenges and making exact learning infeasible beyond small systems. In heterogeneous environments, the approach further struggles to accommodate asymmetries in roles, observation spaces, and information constraints, undermining both robustness and efficiency. These limitations underscore why centralised baselines, while conceptually elegant, are rarely applied in large-scale multi-agent reinforcement learning. Instead, they function primarily as benchmarks, highlighting the need for decentralised or hybrid approaches that can combine tractability with effectiveness in complex domains.

3.1.5 Improving the Centralised Baseline Model

Several strategies have been proposed to overcome the limitations of centralised models, most of which exploit structural properties of the agent population or the environment to reduce the effective complexity of the joint state–action space. One influential line of work focuses on *statistical composition*. Rather than modelling every interaction between agents explicitly, statistical approaches approximate the collective influence of the population through aggregate measures. For instance, **Mean Field Games** [Yang et al., 2018] replace individual pairwise interactions with an average or "mean field" effect, thereby enabling each agent to act as though it interacts with the population distribution rather than with every other agent directly. This framework has been applied successfully in large-scale domains such as traffic optimisation, where each vehicle can respond to the average flow of surrounding vehicles instead of tracking all others. Similarly, techniques of **state and action abstraction** [Li, 2009; Guestrin et al., 2003] compress the problem space by grouping together similar states or actions into equivalence classes, allowing a centralised learner to operate over a reduced but representative model of the environment. In multi-robot navigation, for example, nearby spatial positions may be abstracted into a single state, making learning tractable without losing key structural information.

A complementary set of methods takes advantage of *invariant observations* among homogeneous agents. When agents share identical observation and action spaces, their policies may be parameterised identically, allowing for **parameter sharing** [Gupta et al., 2017]. This technique pools gradients across agents during training, accelerating convergence and improving policy generalisation, and has proven effective in cooperative navigation settings where agents with similar roles must coordinate movement strategies. Extensions of

this idea include the use of **episodic memory mechanisms** [Blumenkamp & Nikolaidis, 2020], which allow agents to store and retrieve successful experiences across episodes. Such memory systems are particularly useful in swarm robotics, where past coordination strategies can be recalled and reused to adapt to new but related tasks.

Together, these advances allow centralised models to scale beyond their naïve formulations, preserving the interpretability and coordination benefits of global decision-making while mitigating the intractability that arises from exponential growth in the joint action–state space. Statistical composition reduces the dimensional burden by abstracting away unnecessary detail, while invariant observation mechanisms exploit homogeneity to ensure learning efficiency and transferability. As a result, improved centralised models provide a more practical foundation for cooperative multi-agent reinforcement learning in complex domains, while continuing to serve as conceptual reference points for evaluating decentralised or hybrid approaches.

3.2 Review of Decentralised Methods

3.2.1 Decentralised Agent Model

The decentralised agent model conceptualises multi-agent systems by decomposing the global decision-making problem into the local perspectives of individual agents. In this framework, each agent selects actions based solely on its own observations, which may differ across agents and fluctuate over time. Unlike centralised models, where full observability of the environment can be assumed, decentralised agents operate under conditions of partial, noisy, and potentially redundant information. This introduces inherent challenges in learning and coordination, as each agent must act optimally without complete knowledge of the global state.

A further complication arises from non-stationarity. Because other agents are simultaneously learning and adapting their policies, the effective environment encountered by any individual agent is continually changing. This non-stationarity distinguishes decentralised settings from traditional single-agent reinforcement learning and necessitates methods capable of stabilising learning despite dynamically evolving interactions. By contrast, in stationary environments, where observations and transition dynamics remain invariant over time, agents can often employ model-free approaches, as the environment can be considered stable from the perspective of each agent.

The influence of other agents, which lie outside an agent's local observation, is typically modelled as part of the environment. When these external agents' actions are unknown, their effects are treated stochastically. To mitigate the resulting uncertainty, decentralised approaches may leverage communication protocols, enabling agents to share information and expand their effective observation space. Alternatively, agents may learn internal models of the environment and other agents, allowing them to infer hidden states or predict the behaviour of others.

Observability represents a critical dimension in decentralised learning. Partial observability generalises the fully observable centralised setting, capturing the uncertainty and informational constraints inherent in distributed systems. From this perspective, decentralised models encompass centralised control as the limiting case of complete global knowledge

while simultaneously representing more realistic scenarios in which agents perceive only a subset of the environment. Recognising this continuum is essential for understanding the trade-offs between computational tractability, coordination, and robustness in multi-agent reinforcement learning.

Overall, the decentralised agent model provides a structured framework for analysing the challenges of learning under uncertainty, limited information, and non-stationary dynamics. It forms the foundation for many contemporary MARL approaches, including independent learners, value function factorisation methods, and communication-enhanced algorithms, all of which aim to improve efficiency, scalability, and coordination in complex multi-agent systems.

3.2.2 Baseline Model for the Decentralised Case

A common baseline in decentralised settings is Independent Q-Learning (IQL) [Tan, 1993; Tampuu et al., 2017], in which each agent treats the other agents as part of the environment and independently learns its own value function. Despite its simplicity, IQL has proven effective in certain cooperative domains, such as grid-based pursuit games [Sen et al., 1994], where agents can achieve implicit coordination through environmental feedback.

// Algorithm controls agent i

Initialize: $Q_i(s, a_i) = 0$ for all $s \in S, a_i \in A_i$
Repeat for every episode:
for $t = 0, 1, 2, \ldots$ **do**
  Observe current state s^t
  With probability ϵ: choose random action $a_i^t \in A_i$
  Otherwise: choose action $a_i^t \in \arg\max_{a_i} Q_i(s^t, a_i)$
  (meanwhile, other agents $j \neq i$ choose their actions a_j^t)
  Observe own reward r_i^t and next state s^{t+1}
  $Q_i(s^t, a_i^t) \leftarrow Q_i(s^t, a_i^t) + \alpha \left[r_i^t + \gamma \max_{a_i'} Q_i(s^{t+1}, a_i') - Q_i(s^t, a_i^t) \right]$

Figure 4:Algorithm 2 Independent Q-Learning (IOL) for stochastic games

3.2.3 Limitations of the Decentralised Baseline Model

Despite its conceptual simplicity and computational tractability, the decentralised baseline exhibits fundamental limitations that constrain its effectiveness in multi-agent settings. A primary challenge arises from the non-stationarity of local observations. Because each agent's environment is influenced by the evolving policies of other agents, the transition dynamics from any single agent's perspective are inherently non-stationary. This instability can prevent convergence in independent learning approaches such as IQL [Tan, 1993]. Even when individual Q-values converge, agents may select incompatible actions, leading to suboptimal joint policies that fail to maximise collective performance [Lauer & Riedmiller, 2000]. Coordination is therefore not guaranteed under purely individual learning schemes.

Another critical limitation is partial and redundant observability. In decentralised frameworks, agents typically have access only to a subset of the full environment state, and this local information may be incomplete or noisy. Furthermore, multiple agents may observe overlapping or redundant aspects of the environment, reducing learning efficiency by diverting computational resources to processing irrelevant data. These limitations hinder

agents' ability to accurately infer the consequences of their actions or predict the behaviour of others, complicating coordination and slowing policy convergence.

Collectively, these challenges underscore why simple decentralised baselines, while useful as reference points, are often insufficient for achieving efficient, scalable, and coordinated behaviour in complex multi-agent reinforcement learning environments. They highlight the need for enhanced strategies that address non-stationarity, improve information sharing, and exploit structural regularities among agents.

3.2.4 Improving the Decentralised Baseline Model

To overcome the limitations inherent in decentralised learning, several advanced strategies have been developed, targeting non-stationarity, partial observability, and coordination challenges in multi-agent systems.

One class of improvements leverages **sparsity and networked solutions** to manage the high-dimensional interaction space among agents. Distributed systems can employ consensus protocols, enabling agents to aggregate and generalise information from a subset of peers, thereby reducing the need for full observability and improving scalability. Similarly, attention mechanisms, as demonstrated in the AlphaStar system [Vinyals et al., 2019], allow agents to selectively focus on the most relevant aspects of their observations, filtering out redundant or irrelevant information and concentrating computational resources on features that most influence coordination.

Coordination and communication mechanisms further enhance decentralised learning. Coordination graphs and networked approaches [Kok and Vlassis, 2006], decompose the joint Q-function into sparse, factorised representations. This enables agents to learn local utility functions while preserving the global interaction structure, reducing the combinatorial complexity of the joint action space. Limited message-passing protocols complement this decomposition by enabling local communication between neighbouring agents during execution, supporting distributed coordination without requiring a fully centralised controller. These mechanisms have been successfully applied in domains ranging from robotic swarms to cooperative game-playing, demonstrating their efficacy under partial observability and dynamic environmental interactions.

Value function factorisation represents another powerful approach, enabling decentralised execution while maintaining joint policy optimality. Methods such as Value Decomposition Networks (VDN) [Sunehag et al., 2017], QMIX [Rashid et al., 2018], and QPLEX [Wang et al., 2021] learn structured decompositions of the global Q-function into per-agent components. By ensuring that local actions collectively approximate the optimal joint policy, these approaches mitigate coordination failures commonly observed in independent learners, particularly in cooperative tasks with tightly coupled objectives.

Optimistic and lenient learning strategies address the instability introduced by non-stationary environments. Distributed Q-Learning [Lauer & Riedmiller, 2000] applies optimistic updates to local Q-values, helping agents adapt when the environment changes due to other agents' learning. Hysteretic Q-Learning [Matignon et al., 2007] introduces stochasticity-aware updates that temper the impact of misleading experiences, while Lenient Q-Learning [Wei & Luke, 2016] combines optimism with selective forgiveness of early mistakes to encourage cooperative behaviour under uncertainty. These strategies have proven

particularly effective in environments where early exploratory actions could otherwise produce persistent suboptimal policies.

Together, these enhancements illustrate that decentralised models can effectively handle non-stationarity, partial observability, and coordination challenges. By incorporating structured communication, networked attention, factorised value functions, and lenient updates, modern decentralised approaches achieve greater stability, coordination, and scalability, enabling efficient multi-agent reinforcement learning in complex cooperative environments. They exemplify the rich spectrum of solutions available to balance autonomy, observability, and global performance in distributed multi-agent systems.

3.3 How to Model Targets and Task Objectives?

3.3.1 Reward Modelling in Multi-Agent Systems

In multi-agent reinforcement learning (MARL), reward modelling defines the objectives that guide agent behaviour, corresponding to subsets of the system's state space that represent desirable outcomes. Just as hierarchical frameworks help interpret agent behaviour, they are equally valuable for conceptualising reward structures. A hierarchical perspective allows for distinguishing between centralised and decentralised reward formulations, providing a structured lens for understanding how objectives are defined and pursued across multiple agents.

In centralised reward models, a single overarching reward function governs all agents. This reward can take cooperative, competitive, or mixed forms. For instance, in cooperative tasks, the centralised reward is often defined at the team level and computed as an aggregation, such as a sum or weighted combination, of individual agent rewards. This approach establishes a direct mapping between individual contributions and collective outcomes, providing a unified measure of task success. In competitive settings, the centralised reward may encode zero-sum dynamics or other game-theoretic objectives, where optimizing one agent's outcome impacts others.

Decentralised reward models, by contrast, assign each agent its own local reward, derived from partial observations or subsets of the overall task objective. While decentralised rewards promote autonomy and scalability, they introduce coordination challenges: agents optimizing locally may fail to achieve globally optimal outcomes, and partial observability can hinder accurate inference of other agents' behaviour.

The choice of reward structure directly informs task definition, the notion of optimality, and the design of learning strategies. By explicitly formalising the reward hierarchy, whether centralised, decentralised, or hybrid, reward modelling provides a conceptual framework for policy design and performance evaluation. Ultimately, it links the hierarchical structure of agents to measurable objectives, guiding the development of effective cooperative or competitive strategies while shaping learning efficiency and agent behaviour in complex multi-agent environments.

3.3.2 Typical Reward Modelling

In multi-agent reinforcement learning, reward modelling typically involves designing functions that provide feedback to agents based on their actions and the resulting state transitions. Common approaches often rely on sparse, discrete, or single-objective reward structures. Sparse rewards provide feedback only upon achieving specific milestones or completing a task, while discrete rewards assign fixed values to outcomes. Single-objective reward models focus on a solitary performance metric, which can simplify policy learning but may inadequately capture the complexity of multi-agent tasks that involve multiple, potentially conflicting objectives.

3.3.3 Limitations in Typical Reward Modelling

Conventional reward modelling faces several well-documented limitations that can hinder learning and coordination. Sparse or poorly aligned rewards can slow convergence or induce unintended behaviours, including reward hacking, where agents exploit loopholes in the reward function without accomplishing the intended task objectives. In decentralized multi-agent settings, these challenges are exacerbated by partial observability: agents typically receive limited or noisy information, making it difficult to infer the global consequences of local actions.

Non-stationarity represents another major limitation in multi-agent systems. Since each agent adapts its policy concurrently, the effective environment from any individual agent's perspective is continually changing, destabilizing learning and preventing reliable convergence in independent learning approaches. Furthermore, as the number of agents increases, centralised reward models that rely on joint states or joint actions may become computationally intractable, whereas decentralised reward structures may fail to ensure coordinated global performance. These factors collectively highlight the difficulty of designing reward functions that are both informative and scalable in complex multi-agent environments.

3.3.4 Limitations in Decentralised Reward Models

Decentralised reward modelling introduces additional challenges. Sparse rewards remain a critical obstacle, as infrequent feedback limits learning efficiency. While reward shaping, providing intermediate or continuous feedback, can accelerate learning, designing shaped rewards that are both informative and aligned with global objectives is non-trivial. Another challenge arises in multi-objective settings, where agents must optimize multiple targets simultaneously. Poorly aligned or redundant objectives can slow learning and coordination, as agents may pursue conflicting local goals that do not contribute effectively to collective performance. These issues underscore the inherent tension between agent autonomy and global task success in decentralised reward formulations.

3.3.5 Limitations in Centralised Reward Models

Centralised reward modelling, in which multiple objectives are combined into a single composite reward, provides a clear pathway for coordinated learning. However, it introduces its own set of challenges. In tasks with multiple objectives, conflicts among goals can create ambiguity in determining the optimal joint policy, potentially resulting in multiple equilibria. Additionally, credit assignment becomes a significant issue: determining the contribution of

individual agents to the global reward is often complex, particularly in heterogeneous systems, complicating both learning and coordination. Finally, the computational burden associated with large joint state–action spaces can limit the practical applicability of fully centralised reward models.

3.3.6 Improving Typical Reward Modelling

Several strategies have been proposed to improve the efficiency and effectiveness of learning through careful reward design in multi-agent reinforcement learning. A key approach is the use of **shaped rewards**, which provide continuous or intermediate feedback rather than relying solely on sparse, terminal signals. Shaped rewards guide agents incrementally toward desirable outcomes, reducing the exploration burden and accelerating convergence. A widely cited method is **potential-based reward shaping** [Ng et al., 1999], in which a potential function defines intermediate rewards proportional to the agent's progress toward the goal. For instance, in robotic manipulation tasks, shaping rewards can be designed based on the negative distance to a target object, effectively incentivizing incremental improvements without altering the optimal policy. Similarly, in navigation tasks, agents can be rewarded for approaching sub-goals, which significantly improves learning speed in complex environments with sparse terminal rewards.

Another strategy involves **multi-objective optimisation**, which addresses scenarios where agents must simultaneously pursue several goals. Instead of combining all objectives into a single scalar reward, multi-objective formulations explicitly model the trade-offs among different targets, allowing agents to balance competing or complementary objectives [Roijers et al., 2013]. For example, in cooperative team games, one objective might maximize individual performance while another ensures equitable contribution among team members. In heterogeneous multi-agent systems, multi-objective reward structures allow agents with distinct roles or capabilities to optimize relevant local targets while contributing to a collective goal. This approach has been successfully applied in domains such as autonomous vehicle coordination, where safety, efficiency, and passenger comfort must be balanced simultaneously.

Designing relevant objectives is another crucial component of improved reward modelling. Aligning intrinsic and extrinsic motivations helps agents explore the state space efficiently while avoiding unproductive behaviours. Distance-based rewards, for example, measure proximity to task goals and encourage agents to minimize the gap between their current and target states. Count-based metrics quantify information acquisition or task completion progress, connecting directly to intrinsic motivation and encouraging agents to reduce uncertainty in the environment. Sequential metrics, which assess the number of steps required to achieve a target, provide temporal guidance and support decision-making in multi-step tasks. Polarity-based metrics, inspired by game-theoretic concepts, can formalize cooperative or competitive incentives, guiding agents to behave in accordance with team or adversarial dynamics.

The formalisation of these metrics often incorporates exponential or decaying functions to balance short-term and long-term incentives, effectively shaping the temporal horizon of learning. In decentralised reward models, where each agent only observes a subset of the environment, carefully designed local reward metrics are critical for enabling coordination

and alignment with global objectives. By integrating shaped rewards, multi-objective optimisation, and well-defined local metrics, reward modelling can significantly improve learning efficiency, facilitate coordination among agents, and ensure that both individual and collective goals are achievable in complex multi-agent reinforcement learning scenarios.

3.4 How to Model Sequence and Decision Process?

3.4.1 Introduction to Sequence Modelling

In multi-agent reinforcement learning, the sequential decision process defines how agents select actions over time and how these actions collectively influence both individual and system-level outcomes. Sequence modelling is critical for understanding temporal dependencies and designing policies that can adapt to dynamically evolving environments. This perspective applies to both centralised and decentralised agent frameworks, with different implications for policy representation, learning efficiency, and coordination.

A conventional baseline approach treats multi-agent interactions as a **simultaneous-move game**, often formulated as a Nash game, where agents select joint actions at each timestep. While this framework provides a theoretically grounded representation of strategic interaction, it suffers from significant practical limitations. The joint action space grows combinatorially with the number of agents, making exploration and learning computationally intractable for all but small systems. Moreover, simultaneous action selection fails to capture natural sequential dependencies inherent in stochastic or temporally extended tasks, which can lead to coordination failures and suboptimal joint performance.

3.4.2 Limitations of Joint Action Sampling

Joint action sampling, where all agents select actions simultaneously—exacerbates several challenges in multi-agent reinforcement learning. First, the **curse of dimensionality** arises because the joint state-action space grows exponentially with the number of agents and the size of their individual action sets. This dramatically increases the sample complexity required for effective learning. Second, **temporal dependencies** among agents' decisions are ignored, which can prevent the emergence of structured strategies or adaptive responses in dynamic environments. For instance, in multi-robot coordination or real-time strategy games, actions taken by one agent often influence subsequent choices of others; ignoring this sequential structure can produce unstable or inefficient policies. Third, in decentralised learning, joint action sampling aggravates **non-stationarity**, as each agent's environment is affected by the evolving policies of all other agents simultaneously, hindering convergence.

3.4.3 Improving on Joint Sampling: Sequential and Auto-Regressive Approaches

To address the limitations of joint-action models, several approaches explicitly incorporate **sequential structure** into multi-agent learning.

Sequential Game Models extend simultaneous-move formulations to include temporal dependencies. For example, **Stackelberg games** introduce a leader-follower hierarchy, where the leader commits to an action first, and followers respond optimally. This framework

reduces combinatorial complexity and enables agents to develop structured strategies, such as aggressive or defensive behaviours, in stochastic environments. Sequential formulations also naturally accommodate hierarchical or temporally extended tasks, where decisions at one timestep condition future actions, improving both coordination and policy interpretability.

Sequential Sampling in Policy Execution leverages the temporal structure of multi-agent interactions to improve exploration and learning efficiency. In practice, methods like **AlphaStar** implement auto-regressive, sequential action selection, where each agent samples actions conditioned on prior agents' choices. This sequential sampling preserves dependencies among agents' actions and mitigates combinatorial explosion in the joint action space. Similarly, **alternating learning schemes** allow agents to update policies sequentially rather than simultaneously, reducing non-stationarity and stabilizing learning in decentralised environments. These approaches demonstrate how sequential modelling can enhance performance in large-scale, cooperative, or competitive multi-agent tasks.

3.4.4 Policy Sampling and Action Selection

The sequential decision process is intimately connected with the design of the **policy sampling mechanism**. Whether agents sample actions stochastically, deterministically, or auto-regressively significantly impacts the balance between exploration and exploitation, the quality of coordination among agents, and overall learning efficiency. Carefully structured sampling enables agents to exploit temporal dependencies, anticipate other agents' behaviour, and make decisions that are robust to stochasticity in the environment.

By explicitly modelling the sequential decision process, researchers can capture temporal correlations, reduce the effective combinatorial complexity of joint action spaces, and design learning algorithms that scale to large multi-agent systems. Sequential modelling not only improves convergence and coordination but also allows policies to be more interpretable and adaptive in dynamic, temporally extended, or hierarchical multi-agent settings.

3.5 Training and Execution Paradigms

3.5.1 Overview of Training and Execution Paradigms

Training and execution paradigms are central to the design and evaluation of learning algorithms in cooperative multi-agent reinforcement learning (MARL). These paradigms define how agents access information, coordinate during learning, and update policies, as well as how the resulting policies are deployed during execution. The choice of paradigm has profound implications for learning efficiency, scalability, coordination, and robustness. Three commonly studied paradigms are centralised training with centralised execution (CTE), decentralised training with decentralised execution (DTE), and centralised training with decentralised execution (CTDE) [Lowe et al., 2017; Rashid et al., 2018]. Each paradigm balances different trade-offs between access to global information, computational complexity, and practical applicability in real-world multi-agent systems.

3.5.2 Training Paradigms

Training can be classified as either centralised or decentralised based on how agents access information and update their policies. In **centralised training**, agents leverage global state and reward information to jointly optimise their policies. This facilitates coordinated behaviour and can improve sample efficiency by allowing agents to learn from the complete system context. Centralised training is particularly useful in environments with tightly coupled objectives or complex dependencies between agents. However, this approach is computationally intensive and becomes increasingly infeasible as the number of agents or the dimensionality of the state and action spaces grows [Foerster et al., 2018].

In contrast, **decentralised training** restricts agents to local observations and rewards. While this promotes scalability and robustness, it introduces significant challenges, including non-stationarity arising from the simultaneous learning of other agents, partial observability of the environment, and difficulties in coordinating joint behaviour [Tan, 1993; Lauer & Riedmiller, 2000]. Despite these challenges, decentralised training more closely mirrors real-world conditions, where agents often operate with limited information and must adapt to evolving dynamics.

Evaluating training performance typically involves metrics such as **policy convergence**, which measures the stability of learned policies across states, **sample complexity**, quantifying the number of interactions needed to achieve satisfactory performance, and **robustness to increasing complexity**, which examines scalability with respect to the number of agents, objectives, or environmental dimensions [Hernandez-Leal et al., 2019]. These metrics provide a systematic basis for comparing learning efficiency across different paradigms.

3.5.3 Execution Paradigms

Execution focuses on how learned policies are deployed to make decisions in the environment. In **centralised execution**, agents act based on global state information, enabling fully coordinated joint actions. While this ensures optimal coordination in principle, it requires full observability at runtime, which may be unrealistic in distributed or partially observable environments.

Decentralised execution, by contrast, allows each agent to select actions independently based on local observations, facilitating scalability and practical deployment in real-world systems. Coordination under decentralised execution must emerge from the learned policies rather than being enforced directly, which can challenge stability and performance but increases adaptability to dynamic and partially observable environments [Oliehoek & Amato, 2016].

Key considerations in execution include the **application of approximate solutions**, where policies learned during training are applied in environments that may differ in complexity or observability, and **greedy policy selection**, often used to evaluate the performance of learned policies without further learning. Metrics such as **steps to optimality**, measuring the speed at which agents reach task objectives, and **error analysis and generalisation**, assessing performance degradation under new or more complex scenarios, provide quantitative measures of execution effectiveness.

3.5.4 Integrating Training and Execution

Distinguishing between training and execution paradigms, and between centralised and decentralised information access, provides a systematic framework for assessing multi-agent learning algorithms. CTDE paradigms, for example, exploit centralised information during training to stabilise learning and then deploy policies in a decentralised manner for practical execution, combining the advantages of both approaches [Lowe et al., 2017]. By carefully selecting the appropriate training and execution paradigm, researchers can balance efficiency, coordination, scalability, and robustness, thereby tailoring MARL systems to the requirements of complex cooperative environments.

3.6 Summary of Analysis on Literature

The literature on cooperative Multi-Agent Reinforcement Learning (MARL) encompasses a range of approaches addressing the core modelling components: agent representations, target formulations, and sequential decision processes. Across these domains, research consistently emphasizes the importance of structuring agent interactions, defining informative objectives, and designing learning sequences that facilitate coordination, efficiency, and scalability. Additional paradigms tackle broader challenges inherent in high-dimensional multi-agent systems, including combinatorial state-action growth, computational complexity, and robustness under partial observability.

A recurring theme in contemporary approaches is the exploitation of **composition, sparsity, and invariance** to mitigate complexity. Invariant properties—features that remain unchanged under specific transformations, serve as powerful tools for simplifying multi-agent systems. Examples include **statistical invariants**, such as the centre of mass, which allow collective representation of multiple agents, and **positional symmetries**, including translational and rotational invariance. These principles have been successfully applied in domains such as computer vision and natural language processing to reduce dimensionality and enhance learning efficiency.

Managing complexity is closely linked to addressing the **curse of dimensionality**, which arises when the joint state-action space grows exponentially with the number of agents or environmental features. Practical approaches include classical statistical methods, such as Principal Component Analysis (PCA), regularization techniques in loss functions, and modern deep learning architectures that automate feature extraction. Theoretical frameworks, including information-theoretic compression, have also been proposed as general strategies for efficient representation and as models of adaptive intelligence [reference].

The relevance of these principles to MARL is threefold. First, they explain why certain state-of-the-art methods achieve success in cooperative environments. Second, they reinterpret typical MARL challenges, such as non-stationarity, sparse rewards, and coordination difficulties, as issues of **complexity management**. Third, they motivate experimental hypotheses exploring efficient modelling, reward shaping, and policy design.

From this review, two overarching strategies emerge for overcoming inefficiencies in cooperative MARL:

1. **Reducing Complexity**: Achieved through sparsity, factorization, or abstraction, which simplify agent interactions and the joint state-action space.
2. **Exploiting Invariants**: Leveraging symmetry, homogeneity, and shared reward structures to generalize learning and facilitate coordinated behaviour across agents.

These principles directly inform the hypotheses and experimental framework of this thesis, guiding the systematic evaluation of efficiency-enhancing modelling, reward design, and sequential decision strategies in cooperative MARL.

4. Methodology

This chapter outlines the methodological framework adopted to investigate efficiency in cooperative Multi-Agent Reinforcement Learning (MARL). The methodology is designed to ensure controlled, reproducible experiments that test the theoretical hypotheses proposed in Chapter 2 and build on insights from the literature review in Chapter 3. Emphasis is placed on scalability, interpretability, and systematic evaluation across key modelling dimensions, including agent representations, reward structures, and sequential decision processes.

4.1 Problem Choice: Cooperative Task Selection

To provide a tractable yet sufficiently challenging environment, this study adopts the Multi-Agent Path Finding (MAPF) task as the primary benchmark. MAPF embodies essential cooperative challenges such as coordination, collision avoidance, and scalability, making it representative of broader cooperative MARL problems [Stern et al., 2019; Ma et al., 2021]. The environment formalizes several key dimensions: the spatial dimension is represented as a discrete two-dimensional grid, and the temporal dimension corresponds to sequential decision points at each timestep. States include the positions of all agents and the current timestep, while agent interactions are captured through local observations of nearby agents, targets, and obstacles, maintaining partial observability to balance informativeness with computational tractability. Agent actions are defined as state-transition vectors, indicating movements from current to subsequent positions. Rewards are scalar, derived from potential-based transitions, task completion, and collision penalties. The MAPF environment supports both centralized and decentralized formulations, enabling systematic assessment of learning efficiency under differing information-sharing paradigms.

4.2 Framework Set-Up and Design Considerations

The experimental framework is structured to investigate efficiency, scalability, and interpretability. Three primary components are considered: agent models, reward models, and sequential decision processes. Each component involves specific design choices intended to facilitate efficient learning and coordinated behaviour.

4.2.1 Agent Models

Agents are modelled either as fully decentralized learners or as partially centralized entities during training. Homogeneous local observations are adopted to simplify analysis while retaining the coordination challenge inherent in multi-agent interactions. Episodic shared histories are integrated to mitigate non-stationarity and improve credit assignment across agents. Design choices include whether to incorporate memory, the granularity of observation sharing, and the degree of centralized information access during learning, each of which directly impacts learning efficiency and stability.

4.2.2 Reward Models

Reward design is critical to guiding cooperative behaviour efficiently. Three key design choices are implemented: sparse rewards delivered upon task completion capture long-term objectives, shaped rewards leverage Artificial Potential Fields with Intrinsic Rewards (APF-IR) to provide continuous, informative signals that accelerate learning while preserving

policy invariance [Ng et al., 1999], and multi-objective rewards encode distance to goals, collision penalties, and exploration incentives. Together, these design choices balance the need for informative feedback with the preservation of global task objectives and facilitate efficient coordination among agents.

4.2.3 Sequential Decision Processes

Sequential decision-making is formalized to capture temporal dependencies among agents. Policies are updated using temporal-difference learning, allowing adaptation to dynamically changing environments. Both simultaneous (joint action) and sequential (Stackelberg-type) formulations are explored to assess the impact of leader-follower dynamics, emergent strategies, and stability under stochastic interactions. Policy sampling strategies, including ε-greedy and information-theoretic approaches, are employed to promote efficient exploration and minimize redundant visits to previously encountered states.

4.3 Learning Methods

Two classes of learning methods are employed. The baseline method, Independent Q-Learning (IQL), treats each agent as learning independently while considering other agents as part of the environment [Tan, 1993]. IQL highlights challenges of non-stationarity, partial observability, and credit assignment in decentralized MARL. The proposed composite method builds upon these insights under a Centralized Training, Decentralized Execution (CTDE) paradigm [Foerster et al., 2018; Lowe et al., 2017], integrating APF-IR reward shaping, shared episodic memory for stability, and information-theoretic exploration to efficiently cover the state space. Centralized training stabilizes learning by leveraging global information, while decentralized execution ensures scalability and real-world applicability. This composite design enables systematic evaluation of the individual contributions of each component while preserving coherence in cooperative MARL tasks.

4.4 Experimental Design

The experimental framework is structured in two complementary phases to evaluate the impact of design choices on cooperative MARL performance.

4.4.1 Theoretical Idealised Experiments

The first phase consists of **conceptual, idealized experiments** designed to isolate the influence of each modelling dimension. These experiments are not implemented in practice but serve as a theoretical framework to understand how individual components affect efficiency, coordination, and learning stability. By fixing other factors at baseline levels while varying a single dimension, the expected contributions of each component can be systematically reasoned. The idealized experiments include:

1. **Reward Modelling:** Theoretically compare sparse rewards provided only at task completion versus shaped rewards based on Artificial Potential Fields (APF), analysing how more informative signals could accelerate learning.

2. **Reward Objectives:** Conceptually evaluate single-target versus multi-target reward structures to assess how the complexity of objectives might influence convergence, policy coordination, and the balance of competing goals.
3. **Reward Types:** Examine the expected effects of extrinsic rewards (e.g., distance-to-goal metrics) versus intrinsic rewards (e.g., exploration or information-gain incentives) on agent exploration and emergent cooperation.
4. **Agent Modelling:** Compare independent learning with only local histories against agents with shared episodic memory to reason about the theoretical mitigation of non-stationarity and improved credit assignment.
5. **Sequential Decision Processes:** Assess sequential (leader-follower) versus simultaneous joint-action strategies to evaluate how sequential dependencies could reduce combinatorial complexity and enhance coordination.

These idealized experiments provide a structured theoretical baseline, enabling predictions about which design choices are likely to contribute most to efficiency and coordination in MARL without incurring implementation overhead.

4.4.2 Phase 2: Practical Composite Experiments

The second phase implements a **composite method** in the MAPF environment, combining multiple dimensions into a realistic cooperative MARL scenario. This includes multiple agents, stochastic transitions, high-dimensional state-action spaces, shaped and multi-objective rewards, shared histories, and sequential sampling strategies. The composite method is compared against the baseline Independent Q-Learning (IQL) approach, allowing evaluation of the cumulative benefits of integrating theoretically informed design choices.

4.5 Evaluation Metrics

Performance is evaluated along multiple axes. Training efficiency is assessed through convergence rate and sample efficiency, while execution efficiency considers cumulative team reward, collision rate, and path optimality. Interpretability is enhanced using visualizations of potential fields, joint value heatmaps, and agent trajectories, which provide insight into emergent coordination patterns.

By clearly separating theoretical exploration (Phase 1) from practical implementation (Phase 2), this methodology allows rigorous reasoning about the potential impact of design decisions and validates their combined effects in realistic cooperative MARL tasks.

5. Results and Findings

This chapter presents the outcomes of the experimental evaluation, comparing the proposed Composite Method with the baseline Independent Q-Learning (IQL) across key dimensions of learning efficiency, stability, and scalability. The findings are interpreted in the context of the theoretical predictions outlined in Chapter 4 and prior literature on cooperative Multi-Agent Reinforcement Learning (MARL).

5.1 Theoretical Predictions

The theoretical analysis suggested that the integration of reward shaping, shared episodic memory, and information-driven exploration would enhance efficiency in cooperative MARL. **Potential-based reward shaping** was hypothesized to provide denser, informative signals that guide agents toward task objectives while preserving policy invariance [Ng et al., 1999]. **Shared episodic memory** was expected to mitigate non-stationarity caused by simultaneously learning agents, stabilizing Q-value updates. **Information-maximization exploration strategies** were anticipated to reduce redundant state visits and encourage exploration of high-uncertainty regions. Collectively, these predictions implied that the Composite Method would outperform IQL in convergence speed, cumulative reward, and learning stability.

5.2 Experimental Findings

The experimental results largely confirm the theoretical predictions. Figures [X] and [Y] illustrate cumulative rewards over training episodes for both methods. The Composite Method consistently achieves higher cumulative rewards and stabilizes more rapidly than IQL, indicating improved convergence and learning stability. In contrast, IQL exhibits slower learning, higher variance, and occasional oscillatory behaviour, reflecting the challenges of non-stationarity and sparse feedback in decentralized MARL.

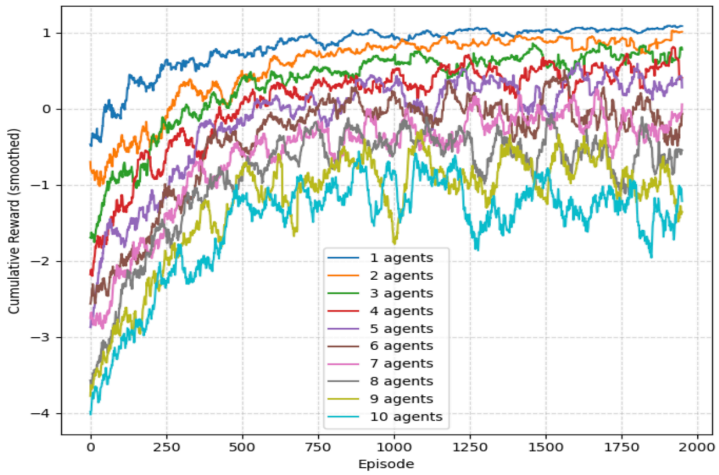

Figure 5: IQL Baseline with Varied Number of Agents

Scaling experiments reveal that increasing the number of agents further enhances the convergence rate of the Composite Method. This suggests that its modular, invariant-aware design effectively exploits shared structural information across high-dimensional joint state spaces. **Sample efficiency** is similarly improved; agents trained under the Composite Method

require fewer interactions to reach stable policies, as reward shaping and information-driven exploration reduce redundant sampling of low-value states.

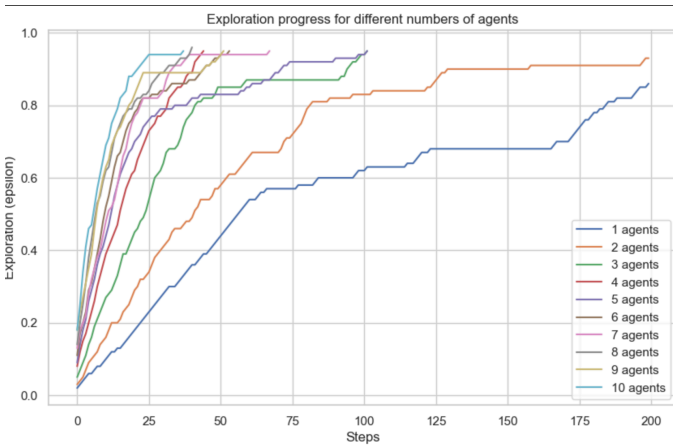

Figure 6: Composite Approach with Varied Number of Agents

Execution efficiency also benefits from the proposed design. Policies learned via the Composite Method achieve higher task success rates, improved path efficiency, and reduced collisions, while maintaining comparable per-episode execution time to IQL. This demonstrates that the observed improvements derive from more efficient learning rather than additional computational overhead during deployment.

5.3 Limitations and Considerations

Several limitations of the experimental evaluation warrant discussion. The observed improvements result from the combination of reward shaping, shared memory, and directed exploration, making it challenging to attribute performance gains to individual components. Differences in update rules and learning assumptions between the Composite Method and IQL further complicate direct comparisons, particularly as agent populations scale. Additionally, variations in state representation and encoding may have contributed to some observed efficiency gains, highlighting the need for controlled isolated studies to see the impact of specific components.

5.4 Summary of Findings

In summary, the Composite Method demonstrates significant improvements over IQL in cooperative MARL. It accelerates convergence, increases cumulative reward, enhances sample efficiency, and stabilizes learning, while maintaining robust execution performance. The integration of reward shaping shared episodic memory, and information-driven exploration facilitates coordination among agents and scales effectively with increasing agent numbers. These findings underscore the potential of modular, invariant-based designs for real-world cooperative multi-agent systems. Future work should focus on dissecting the contributions of individual components, exploring alternative reward structures and exploration strategies, and extending evaluation to more heterogeneous and complex MARL environments.

6. Discussion

This chapter integrates theoretical predictions with experimental findings, focusing on agent observations, target structures, and sequential decision processes in cooperative Multi-Agent Reinforcement Learning (MARL). Practical implications, limitations, and future directions are also considered.

6.1 Theoretical Insights

6.1.1 Agent Observations

Three paradigms were considered: centralized, decentralized, and hierarchical. Centralized observations provide full environmental information, yielding idealized coordination benchmarks but are often impractical. Decentralized observations rely on local perceptions, facing challenges from partial observability and non-stationarity. Hierarchical models, combining local and aggregated high-level information, theoretically stabilize learning and improve convergence while remaining more generalizable.

6.1.2 Target Observations

Centralized reward signals theoretically improve coordination by providing consistent global feedback, while decentralized targets introduce credit assignment challenges. Multi-objective or hierarchical target structures are predicted to facilitate alignment between local and global goals.

6.1.3 Sequential Decision Processes

Simultaneous joint-action models are vulnerable to combinatorial explosion and oscillatory learning. Sequential or leader-follower models reduce effective joint action complexity, mitigate non-stationarity, and support structured coordination.

6.2 Practical Findings

6.2.1 Agent Observations

Experimental results confirmed theoretical trends. Centralized observations accelerate convergence but are unrealistic in occluded or partially observable environments. Decentralized agents benefit from shared episodic memory, which mitigates non-stationarity and improves credit assignment. Hierarchical observation structures, though not fully implemented, are likely to further enhance generalization and execution efficiency.

6.2.2 Target Observations

Reward shaping using potential-based approaches improved convergence under decentralized targets, providing denser intermediate feedback. Multi-objective rewards enhanced coordination but required careful parameter tuning. Centralized rewards consistently yielded faster learning, supporting theoretical predictions.

6.2.3 Sequential Decision Processes

Sequential decision processes stabilized learning, reduced non-stationarity, and improved coordination, particularly in dense or high-dimensional joint state spaces. Simultaneous action selection occasionally led to suboptimal behaviour under collision constraints. Structured exploration and sequential updates were critical for efficient policy learning.

6.3 Methodological Limitations

Limitations include artifacts from discrete grid-world representations, which reduced feedback granularity near collisions, and dependence on extensive state-space coverage for successful execution. Default values for unexplored states occasionally resulted in suboptimal decisions. Continuous state representations, interpolated value estimation, or model-based inference could improve robustness.

7. Conclusion

7.1 Theoretical Insights

This dissertation investigated efficiency in cooperative Multi-Agent Reinforcement Learning (MARL) through analysis of agent models, reward structures, and sequential decision processes. Key challenges identified include non-stationarity, partial observability, credit assignment, and the combinatorial growth of joint state-action spaces. Centralised, decentralised, and hierarchical frameworks were examined, showing that exploiting structural invariants—such as symmetry, homogeneity, and statistical regularities—can reduce complexity and enhance learning efficiency. These insights informed the design of the Composite Method.

7.2 Practical Findings

Experimental evaluation on the Multi-Agent Path Finding (MAPF) benchmark demonstrated that the Composite Method outperforms baseline Independent Q-Learning (IQL) in training efficiency, convergence speed, and coordination. Improvements were achieved through reward shaping with artificial potential fields, shared episodic memory, and information-maximization exploration. Limitations included artifacts from discrete state representations, sensitivity to incomplete state-space coverage, and difficulty isolating contributions of individual components, highlighting trade-offs between theoretical clarity and practical implementation.

7.3 Future Directions

Future work should focus on:

1. **Exploiting Invariants for Efficiency:** Developing abstractions and invariant representations to further reduce state-action complexity and improve learning scalability.
2. **Enhancing Interpretability:** Visualizing emergent coordination, reward shaping, and policy evolution to better understand agent behaviour and support deployment in real-world systems.
3. **Extending to Continuous and Realistic Environments:** Validating methods in continuous spaces, robotic platforms, and networked systems to ensure generality beyond discrete benchmarks.

In summary, this study demonstrates that integrating theoretical analysis with principled experimental design can substantially improve efficiency in cooperative MARL. The work provides a systematic framework linking formal models, reward design, and sequential decision-making to practical strategies for scalable, coordinated multi-agent learning.

Appendix One

The code and experimental resources supporting this thesis are available in the GitLab repository: https://git.cs.bham.ac.uk/projects-2024-25/cxw427. The repository contains four Jupyter notebooks, with the primary notebooks implementing the Composite Method and the Independent Q-Learning (IQL) baseline. These notebooks encompass data preprocessing, environment setup for the Multi-Agent Path Finding (MAPF) task, reward shaping mechanisms, sequential decision processes, and evaluation metrics. The repository provides a complete, reproducible framework for training and evaluating cooperative MARL algorithms as described in this thesis.

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